Quality of life and unmet needs in patients with chronic liver disease: A mixed method systematic review

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Table S1. Search strategy

<table>
<tr><th colspan="3">Patients´ experience of having a liver disease and the impact on their quality of life</th></tr>
<tr><td colspan="3">Search terms: Quality of life, lived experience, patient experience, perception, impact, attitude, behaviour, qualitative research, qualitative study, liver disease, chronic liver disease, cirrhosis, autoimmune liver disease, cholestatic liver disease, viral hepatitis, hepatitis B, hepatitis C, non-alcoholic fatty liver disease, “, hepatocellular carcinoma</td></tr>
<tr><td colspan="2">Criteria´s:
Inclusion Exclusion</td><td>Databases:</td></tr>
<tr><td>Study design:
Qualitative studies
Quantitative studies such as cross-sectional studies, prosepctive studies, and case-control studies
Language:
Danish or English
Limits:
Human
Period 2000-2020
Patient population:
Patients with liver disease
Phenomena of interest:
Experience of liver diseases and the impact on quality of life using interview or</td><td>Study design
Randomised controlled trials
Intervention studies
Studies investering the experience or quality of life of healthcare professionals or relatives
Studies comparing patients with liver disease with patients with other diseases</td><td>Databases:
Cinahl (via EBSCO)
Embase (via Ovid)
PubMed (via Ovid)
Manual search:
Reference list of all articles selected for critical appraisal</td></tr>
</table>

generic or liver specific questionnaires • Control groups or general population for comparison		

Example of search in PubMed (via Ovid)

Search ID Terms	Search Terms
S1	(Quality of life) OR (Mesh "Quality of life")
S2	(Experience) OR ("Lived experience") OR ("Patient experience")
S3	(Perception) OR (Mesh "Perception")
S4	(Impact)
S5	(Attitude) OR (Mesh "Attitude")
S6	(Behavior) OR (Mesh "Behavior")
S7	(Qualitative research) OR (Mesh "Qualitative research")
S8	(Qualitative study) OR (Mesh "Qualitative study")
S9	S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8
S10	(Liver disease) OR (Mesh "Liver disease")
S11	(Chronic liver disease)
S12	(Cirrhosis) OR (Liver cirrhosis) OR (Mesh "Liver cirrhosis")
S13	(Autoimmune liver disease) OR (MESH "Autoimmune liver disease")
S14	("Cholestatic liver disease") OR (Mesh "Cholestatic liver disease")
S15	("Viral hepatitis") OR (Mesh "Viral hepatitis")
S16	("Hepatitis B") OR ("Hepatitis C")
S17	("Non-alcoholic fatty liver disease") OR (NAFLD) OR (Mesh "Non-alcoholic fatty liver disease")

S18	(“Non-alcoholic steatohepatitis”) OR (NASH) OR (Mesh “Non-alcoholic steatohepatitis”)
S19	(“Hepatocellular carcinoma”) OR (HCC) OR (Mesh “Hepatocellular carcinoma”)
S20	S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19
S16	S9 AND S20
Search retrieved 1.444 hits Conducted on 20th March 2020	
Legend Mesh – Medical Subject Heading / MeSH term ( ) – Keywords will be searched in this order: MeSh, Journals, Authors **Limits** Danish or English language, Human, Period 2000-2020	

Table S2. Reason for exclusion of full-text articles

Authors	Exclusion reasons
Afendy et al 2009	Etiology of liver disease not described
Che et al 2014	Extraction of disease group not possible
Fan et al 2010	Systematic review
Frith et al 2010	Investigate falls not quality of life
Gao et al 2013	Extraction of cirrhosis patients not possible
Jacoby et al 2005	Data extraction not possible
Mazzone et al 2013	Investigate quality of life in mothers´ to children with NAFLD
Kaiser et al 2014	Investigate pain not quality of life
Kim et al 2018	Extraction of cirrhosis patients not possible
Labenz et al 2019	Comparison between groups of patients with cirrhosis
Román et al 2013	Data extraction not possible
Sawamoto et al 2016	Data extraction not possible
Selmi et al 2007	Data extraction not possible
Solá et al 2012	Data extraction not possible
Stepanova et al 2018	Data extraction not possible
Van der Plas et al 2007	Extraction of different disease groups not possible
Van Le et al	Investigate stigma and discrimination not quality of life
Van Os et 2007	Investigate depression not quality of life
Younossi et al 2019	Same study population as in other included study
Zickmund et al (2003)	Data extraction not possible

Afendy A, Kallman JB, Stepanova M et al. Predictors of health-related quality of life in patients with chronic liver disease. Aliment Pharmacol Ther 2009; 30:469-476.

Che Y, You J, Chongsuvivatwong V et al. Dynamics and liver disease specific aspects of quality of life among patients with chrnoc liver disease in Yunnan, China. Asian Pac J Cancer Prev 2014; 15:4765-4771.

Fan et al. Health-related quality of life in patients with hepatocellular carcinoma: a systematic review. Clin Gastroenterol Hepatol 2010; 8:559-564.

Frith J, Kerr S, Robinson L et al. Primary biliary cirrhosis is associated with falls and significant fall related injury. Q J Med 2010; 103:153-161.

Gao F, Fao R, Guang L et al. Health-related quality of life and survival in Chinese patients with chronic liver disease. Health Qual Life Outcomes 2013, 11:131-139.

Jacoby A, Rannard A, Buck D et al. Development, validation, and evaluation of the PBC-40, a disease specific health related quality of life measure for primary biliary cirrhosis. Gut 2005; 54:1622-1629.

Kaiser et al. Important and relevant symptoms including pain concerns in hepatocellular carcinoma (HC): a patient interview study. Support Care Cancer 2014; 22:919-926.

Kim HJ, Chu H, Lee s. Factors influencing on health –related quality of life in South Korean with chronic liver disease. Health Qual Life Outcomes 2018; 16:142-150.

Labenz C, Toenges G, Schattenberg JM et al. Health-related quality of life in patients with compensated and decompensated liver cirrhosis. Eur J Intern Med 2019; 70:54-59.

Mazzone L, Postorino V, De Peppo L et al. Paediatric non-alcoholic fatty liver disease: impact on patients and mothers´ quality of life. Hepat Mon 2013; 13:1-8.

Román E, Córdoba J, Torrens M et al. Falls and cognitive dysfunction impair health-related quality of life in patients with cirrhosis. Eur J Gastroenterol Hepatol 2013; 25:77-84.

Sawamoto R Nagano J, Kajiwara E et al. Inhibition of emotional needs and emotional wellbeing predict disease progression of chronic hepatitis C patients: an 8-year prospective study. Biopsychosoc Med 2016; 24:1-10.

Selmi C, Gershwin ME, Lindor KD et al. Quality of life and everyday activities in patients with primary biliary cirrhosis. Hepatology 2007; 46:1836-1843.

Solá E, Watson H, Graupera I et al. Factors related to quality of life in patients with cirrhosis and ascites: relevance of serum sodium concentration and leg edema. J Hepatol 2012; 57:1199-1206.

Stepanova M, Younossi I, Racila A et al. Prediction of health utility scores in patients with chronic hepatitis C using the chronic liver disease questionnaire-hepatitis C version (CLDQ-HCV). Value Health 2018; 21:612-621.

Van der Plas SM, Hansen BE, de Boer, JB et al. Generic and disease-specific health related quality of life of liver patients with various aetiologies: a survey. Qual Life Res 2007; 16:375-388

Van Le T, Vu TTM, Mai HT et al. Social determinants of stigma and discrimination in Vietnamese patients with chronic hepatitis B. Int J Environ Res Public Health 2019; 16:398-407.

Van Os E, van den Broek WW, Mulder PGH et al 2007. Depression in patients with primary biliary cirrhoisis and primary sclerosing cholangitis. J Hepatology 2007; 46:1099-1103.

Wainwright SP. Transcending chronic liver disease: a qualitative study. J Clin Nurs 1997; 6:43-53.

Younossi ZM, Stepanova M, Lawitz EJ et al. Patients with nonalcoholic steatohepatitis experience severe impairment of health-related quality of life. Am J Gastroenterol 2019; 114:1636-1641.

Zickmund S, HO EY, Masuda M et al. They treated me like a leper. J Gen Intern Med 2003; 18:835-844.

Table S3. Appraisal instrument from Joanna Briggs Institute (JBI)

JBI Critical Appraisal Checklist for Qualitative Research

	Yes	No	Unclear	Not applicable
1. Is there congruity between the states philosophical perspective and the research methodology?	□	□	□	□
2. Is there congruity between the research methodology and the research question or objectives?	□	□	□	□
3. Is there congruity between the research methodology and the methods used to collect data?	□	□	□	□
4. Is there congruity between the research methodology and the representation and analysis of data?	□	□	□	□
5. Is there congruity between the research methodology and the interpretation of results?	□	□	□	□
6. Is there a statement locating the researcher culturally or theoretically?	□	□	□	□
7. Is the influence of the researcher on the research, and vice-versa, addressed?	□	□	□	□
8. Are participants, and their voices, adequately represented?	□	□	□	□
9. Is the research ethical according to current criteria or, for recent studies, and is there evidence of ethical approval by appropriate body?	□	□	□	□
10. Do the conclusions drawn on the research report flow from the analysis, or interpretation, of the data?	□	□	□	□

JBI Critical Appraisal Checklist for Quantitative Research (Checklist for cross-sectional studies with one additional question (no. 3) from checklist for case-control studies)

	Yes	No	Unclear	Not applicable
1. Were the criteria for inclusion in the sample clearly defined?	□	□	□	□
2. Were the study subjects and the setting described in detail?	□	□	□	□
3.Were the study subjects consecutive included?	□	□	□	□
4. Was the exposure measured in a valid and reliable way?	□	□	□	□
5. Were objective, standard criteria used for measurement of the condition?	□	□	□	□
6. Were confounding factors identified?	□	□	□	□

7. Were strategies to deal with confounding factors stated?	□ □ □ □
8. Were outcomes measured in a valid and reliable way?	□ □ □ □
9. Was appropriate statistical analysis used?	□ □ □ □

Table S4. Data extraction from the included studies

Author(s)	Year	Type of study	Methodology	Number of participants	Characteristics of participants	Phenomena of interest	Outcome	Authos(s) conclusion	Quality assessment
Autoimmune liver disease									
Blackburn et al (England)	2007	Quantitative study (questionnaire)	Cross-sectional	24 with PBC	PBC: Mean age 58 years, 42% with cirrhosis	QOL using primary biliary cholangitis questionnaire (PBC-40)	Patients marked fatigue as well as social and emotional domains highest	Patients with PBC has impaired QOL, especially those with fatigue. PBC may affect patients social life negatively	6
Sogolow et al (USA)	2008	Quantitative study (questionnaire)	Case-control	80 with PBC GP from population norms from questionnaire	PBC: 100% woman, 6% under 40 years, 19% between 41-50 years, 42% between 51-60 years, 31% over 61 years	QOL using short-form 36 (SF-36)	Patients with PBC had worse physical and social functioning, role limitation due to emotional and physical health problems, general and mental health, bodily pain and vitality than GP	Patients with PBC has worse QOL than GP. Fatigue was associated with QOL	7
Gulati et al (USA)	2013	Quantitative study (questionnaire)	Cross-sectional	30 children with AILD, AIH in 40% ,	AILD: 43% boys, mean age 12 years.	QOL using pediatric quality	Mean overall QOL score 71.6 in AILD	QOL is lower in children with AILD compared to GP,	7

				PSC in 45%, AIH/PSC in 15% GP from population norms from questionnaire	GP: 52% boys, mean age 8 years	of life inventory (PedsQL)	children and 83.9 in control children	especially with frequent liver disease-related symptoms, even in early stages of disease	
Schramm et al (Germany)	2014	Quantitative study (questionnaire)	Prospective study	103 with AIH GP from population norms from questionnaire	AIH: 29% men, mean age 50 years, 27% with cirrhosis. GP not described.	QOL using short-form 12 (SF-12)	Physical component score in patients with AIH did not differ from GP. Mental component score was reduced in patients with AIH compared to GP	Patients with AIH has reduced mental wellbeing, but non-impaired physical wellbeing compared with GP	9
Untas et al (France)	2015	Quantitative study (questionnaire)	Cross-sectional	130 with PBC	PBC: 100% woman, mean age 57 years.	QOL using short-form 12 (SF-12)	Physical component score 39.8, mental component score 40.1	Woman with PBC has poor QOL	7
Raszeja-Wyszomirska et al (Poland)	2015	Quantitative study (questionnaire)	Cross-sectional	102 with PSC GP from population	PSC: 73% men, median age 35 years, 30% with cirrhosis.	QOL using short-form 36 (SF-36)	Patients with PSC had worse physical and social functioning, role	QOL in patients with PSC is impaired	8

				norms from questionnaire	GP not described.		limitation due to physical health problems, general health perceptions, bodily pain, and vitality than GP.		
Raszeja-Wyszomirska (Poland)	2015	Quantitative study (questionnaire)	Cross-sectional	33 with PBC	PBC: 67% men, mean age 35 years, 21% with cirrhosis.	QOL using short-form 36 (SF-36) and primary biliary cholangitis questionnaire (PBC-27 and PBC-40)	In SF-36 Patients marked the lowest scoring in general health. In both PBC-27 and PBC-40 patients marked fatigue as well as social and emotional domains	Patients with PBC has low QOL. Fatigue, social and emotional domains may affect QOL	6
Dyson et al (England)	2016	Quantitative study (questionnaire)	Prospective study	1990 with PBC	PBC: 9% men, median age 65 years.	QOL using self-made primary biliary cholangitis quality of life question (PBC-QoL)	66% of patients reported good/neutral score and 34% reported poor score	Majority of patients has no impaired QOL. Age at presentation, symptoms, and social dysfunction contributes to QOL impairment	7

Cheung et al (Canada)	2016	Mixed-method study (questionnaire)	Cross-sectional	99 with PSC GP from population norms from questionnaire	51% men, mean age 46 years, 48% with cirrhosis. GP not described.	QOL using short-form 36 (SF-36)	Patients with PSC had worse physical and social functioning, role limitation due to physical health problems, general health perceptions, mental health, and vitality than GP. 75% expressed anxiety and 25% disclosing social isolation	Patients with PSC have lower QOL than GP. PSC lead to a significant psychologic burden that was expressed by existential anxieties and social isolation. Patients highlighted the need for emotional and metal support, and a desire for more information on PSC	8
Raszeja-Wyszomirska (Poland)	2016	Quantitative study (questionnaire)	Case-control	205 with PBC GP from population norms from questionnaire	PBC: 8% men, mean age 58 years. GP not described.	QOL using short-form 36 (SF-36)	Patients with PBC had worse physical and social functioning, role limitation due to physical and emotional health problems, general health perceptions,	Quality of life is significantly impaired in patients with PBC, independently of gender and disease severity	8

							bodily pain and vitality than GP		
Takahashi et al (Japan)	2018	Quantitative study (questionnaire)	Cross-sectional survey	265 with AIH 97 healthy controls	AIH: 13% men, median age 65 years, 11% with cirrhosis. Healthy controls: 20% men, median age 57 years	QOL using chronic liver disease questionnaire (CLDQ) and short-form 36 (SF-36)	Mean overall QOL score 5.5 in AIH and 6.2 in healthy controls. Patients with AIH had worse physical and social functioning, role limitation due to physical and emotional health problems, general health perceptions, bodily pain and vitality than health controls	Patients with AIH showed impaired QOL, which was associated with disease progression, comorbid diseases and treatment.	8
Trevizoli et al (Brazil)	2018	Quantitative study (questionnaire)	Cross-sectional	43 children with AIH 62 control children	AIH: 35% boys, mean age 15 years. Control children: 45% boys, mean age 13 years.	QOL using pediatric quality of life inventory (PedsQL)	Mean overall QOL score 77.9 in AIH children and 84.4 in control children	AIH has a negative effect on QOL among children. The presence of symptoms and dislike of taking medication were	7

								associated with worsening of QOL	
Yagi et al (Japan)	2018	Quantitative study (questionnaire)	Cross-sectional	496 with PBC	PBC: 11% men, mean age 66 years	QOL using short-form 36 (SF-36) and primary biliary cholangitis questionnaire (PBC-40)	Highest domain in PBC-40 emotional. 42.6 physical functioning, 43.4 role-physical, 50.9 bodily pain, 44.5 general health perception, 48.1 vitality 47.1 social functioning, 45.3 role-emotional, 49.9 mental health	QOL impaired in patients with PBC, especially in physical and role-social components	6
Wong et al (England)	2018	Quantitative study (questionnaire)	Cross-sectional	990 with AIH GP from population norms from questionnaire	AIH: 20% men, mean age 58 years, 33% with cirrhosis. GP not described.	QOL using the European quality of life 5 dimension 5 level (EQ-5D-5L)	Mean median EQ-5D-5L UI value was 0.89, and the median EQ-VAS was 80. Values was significant lower than the GP	QOL is impaired in patients with AIH. Treatment with corticosteroid was associated with decreased QOL.	7
Bozzini et al (Brazil)	2019	Quantitative study (questionnaire)	Cross-sectional	80 children with AIH 45 healthy controls	AIH: 30% boys, mean age 13 years.	QOL using pediatric quality of life inventory (PedsQL)	Mean overall QOL score 78.25 in AIH children and 85.63 in control children	Reduced scores in the physical, emotional, and school domains	6

					Healthy controls: Mean age 14 years.			were observed in children with AIH compared to healthy controls. Abdominal pain and corticosteroid treatment negatively affected QOL	
Janik et al (Poland)	2019	Quantitative study (questionnaire)	Cross-sectional	140 with AIH 170 healthy controls	AIH: 23% men, mean age 40 years, 55% with cirrhosis. MELD mean 9.6. Health controls: 30% men, mean age 37 years.	QOL using short-form 36 (SF-36)	Patients with AIH had worse physical and social functioning, role limitation due to physical problems, general health perceptions, bodily pain and vitality than GP	QOL is impaired in patients with AIH.	8
Milovanovic et al (Serbia)	2020	Quantitative study (questionnaire)	Cross-sectional	130 with PBC	PBC: 4% men, mean age 59 years.	QOL using primary biliary cholangitis questionnaire (PBC-27 and PBC-40)	Mean total score for PBC-27 was 60.5 and the mean total PBC-40 score was 89.4	PBC have significant impairment in QOL with fatigue being the most prevalent symptom.	8

								Emotional and social domains are also affected	
Jorgensen et al (USA)	2006	Qualitative study	Phenomenological	8 with PBC	PBC: 12% men, age between 38-60 years	Semistructured interview	Five themes were identified: An unreliable body, fatigue as always present yet insidious, planning a life to conserve energy, struggling to maintain normality and emotional consequences	It is important for healthcare providers to understand the nature of fatigue and appreciate how it differs from the explainable tiredness common to the healthy people	7
Montali et al (Italy)	2011	Qualitative study	Phenomenological	23 with PBC	PBC: 100% women, mean age 59 years,	Semistructured interview	The delegitimation of woman´s experience, PBC as a challenge to woman´s social identity, and the need for biographical continuity.	Patients´ physical and emotional well-being risk damage, as they end up forsaking legitimate reappropriation and reclamation of their feelings and needs	7

Hepatitis									
Bonkovsky et al (USA)	2007	Quantitative study	Prospective study	1144 with hepatitis C GP from population sample	Hepatitis C: 72% men, mean age 50 years GP 53% men, mean age 40 years	QOL using Short-from 36 (SF-36)	Patients with hepatitis C had lower score in all domains compared to GP	QOL is decreased in patients with hepatitis C	7
Björnsson et al (Sweden)	2009	Quantitative study	Cross-sectional	158 with hepatitis C GP from population sample	Hepatitis C: 62% men, mean age 46 years GP not described	QOL using European Quality of Life scale (EQ-5D) and Short-from 36 (SF-36)	Mean EQ-5D score was 0.81 in hepatitis C and 0.81 in GP Mental and physical component score for hepatitis C 43 and 49. For GP 50 and 49	Impairment of QOL was associated with the severity of liver disease	7
Lam et al (China)	2009	Quantitative study	Cross-sectional	520 with hepatitis B GP from population sample	Hepatitis B: 74% men, mean age 50 years, 33% with cirrhosis GP: 47% men	QOL using Short-Form Six-Dimension (SF-6D), Short-Form 36 (SF-36) and Chronic Liver Disease	SF-6D 0.73, mental and physical component score 49.0 and 43.7, mean overall CLDQ score 5.6	QOL was significantly lower than of the general population. Disease severity, treatment, comorbidities, younger age and	7

						questionnaire (CLDQ)		female gender was also associated with poor QOL	
Haq et al (Pakistan)	2012	Quantitative study	Cross-sectional	390 with hepatitis B	Hepatitis B: 58% men, mean age 39 years	QOL using European Quality of Life scale (EQ-5D)	Mean EQ-5D score was 0.37 and EQ-VAS score was 57.12	Hepatitis B has an adverse effect on patients´ QOL. Higher education emerged as a positive predictor of QOL	7
Abdo (Saudi Arabia)	2012	Quantitative study	Prospective study	220 with hepatitis B or C	Hepatitis B or C: 46% men, men age 42 years, 7% with cirrhosis	QOL using Short-Form 36 (SF-36)	Physical and social functioning 66.8 and 78.3, role limitation due to physical and emotional problems 58.6 and 62.1, energy 55.4, well-being 68.1, pain 71.9, general health 60.5	QOL in patients with hepatitis B or C is impaired, especially in those with hepatitis C	7
Woo et al (Canada)	2012	Quantitative study	Cross-sectional	294 with hepatitis B GP from population	Hepatitis B: 63% men, mean age 47 years	QOL using European Quality of Life scale (EQ-	Mean EQ-5D score was 0.92 and EQ-VAS score was 80.	QOL in patients with hepatitis B is only impaired in	6

				norms from questionnaire	GP not described	5D) and short-from 36 (SF-36)	Physical component summary score 53.1 and mental component summary score 50.1	the later stages of liver disease	
Drazic et al (Australia)	2013	Quantitative study	Cross-sectional	77 with hepatitis B or C	Hepatitis B and C: 58% woman, mean age 48 years	QOL using chronic liver disease questionnaire (CLDQ)	Mean CLDQ was 4.10	The results suggest that improved information and care is needed to improve QOL	6
Zhou et al (China)	2013	Quantitative study	Cross-sectional	460 with hepatitis B	Hepatitis B: 66% men, mean age 34 years, 30% with cirrhosis	QOL using chronic liver disease questionnaire (CLDQ)	Mean CLDQ was 5.7	QOL is impaired in patients with hepatitis	8
Strauss et al (Brazil)	2014	Quantitative study	Prospective study	67 with hepatitis B or C 67 Controls	Hepatitis B and C: 68.7% and 57.1% men, mean age 36.2 years. Controls age, race, and sex matched	QOL using short-form 36 (SF-36)	QOL in patients with hepatitis B and C was lower than controls	Decrease in QOL was detected in patients in the early stages of hepatitis B and C	6

Zhuang et al (China)	2014	Quantitative study	Cross-sectional survey	460 with hepatitis B	Hepatitis B: 66% men, mean age 36 years, 30% with cirrhosis	QOL using SF-36, SF-6D, and CLDQ	Physical component score 49.5, mental component score 42.0 in hepatitis B. Physical component score 55.0, mental component score 48.3 in controls 0.68 in hepatitis B and 0.775 in controls in SF-6D. Mean CLDQ score 5.2	Hepatitis B patients suffered significant impaired QOL compared to controls	8
Abedi et al (Iran)	2015	Quantitative study	Cross-sectional	210 with hepatitis B	Hepatitis B: 56% woman	QOL using chronic liver disease questionnaire (CLDQ) and World Health Organization questionnaire (WHOQOL-BREF)	In CLDQ: Abdominal symptoms 2.20, activity 5.54, fatigue 2.22, emotional function 2.64, worry 2.36. In WHOQOL-BREF: Physical 2.79, mental 2.81, social	QOL in patients was below average and social relationship and emotional function was affected	4

							2.80, environmental 2.74		
Mittal et al (India)	2015	Quantitative study	Prospective study	35 with hepatitis B or C 30 controls	Hepatitis B or C: 74% woman, mean age 32 years. Controls: 37% woman, mean age 32 years	QOL using the World Health Organization Quality of Life-Bref (WHO QoL-Bref) and SF-36	QOL scores was lower in patients with hepatitis compared to controls	QOL was decreased in patients with hepatitis compared to controls	6
Kim et al (Korea)	2015	Quantitative study	Cross-sectional survey	7098 with hepatitis B 35490 GP	Hepatitis B: 56% men, age between 20 to > 60 years. GP: 56% men, age between 20 to > 60 years	QOL using European Questionnaire (EQ-5D)	EQ-5D score was 93.1 and EQ-VAS score was 70.2 in hepatitis B and EQ-5D score was 93.8 and EQ-VAS 71.2 in GP	Hepatitis B patients scored lower in QOL than GP	8
Karacaer et al (Turkey)	2016	Quantitative study	Cross-sectional	4257 with hepatitis B	Hepatitis B: 60% men, mean age 40 years	QOL using Short-form 36 (SF-36) and Hepatitis B Quality of Life (HBQOL)	Physical component score 75.5, mental component score 70.6 HBQOL total score 81.4	QOL was higher than expected , but can be worsen when the disease becomes active	7

Doyle et al (Australia)	2016	Quantitative study	Prospective study	111 with hepatitis C	Hepatitis C: 71% men, mean age 34 years	QOL using Short-form 12 (SF-12)	Physical component score 54, mental component score 46	QOL may be impaired by ongoing treatment	7
Liu et al (China)	2016	Quantitative study	Cross-sectional survey	519 with hepatitis B	Hepatitis B: 58% men, mean age 39 years, 22% with cirrhosis	QOL using chronic liver disease questionnaire (CLDQ)	Mean CLDQ was 5.2	QOL in patients was impaired, especially in those with female gender, disease severity, long time absence, no job or retirement	7
Vahidnia et al (USA)	2017	Quantitative study	Cross-sectional	209 with hepatitis B 315 with hepatitis C 1565 controls	Hepatitis B: 65% men, mean age 38 years. Hepatitis C: 59% men, mean age 45 years. Controls: 48% men, mean age 42 years	OQL using European Questionnaire Five Dimensions (EQ-5D)	Mean EQ-5D score 0.93 and EQ-VAS 85.7 for hepatitis B. Mean EQ-5D score 0.83 and EQ-VAS 77.8 for hepatitis C. Mean EQ-5D score 0.94 and EQ-VAS 87.6 for controls.	QOL is impaired in patients with hepatitis compared to controls, especially those with hepatitis C	7

Liu et al (China)	2017	Quantitative study	Cross-sectional	634 with hepatitis B	Hepatitis B: 72% men, mean age 48 years, 29% with cirrhosis	QOL using Short-Form 12 (SF-12)	Physical component summary score 46.2 and mental component summary score 49.1	QOL is impaired in patients with hepatitis	6
Schwarzenberg et al (USA and Canada)	2017	Quantitative study	Prospective study	244 children with hepatitis B GP from population sample	Hepatitis B: 61% girls, mean age 12 years GP: Not described	QOL using the Child Health Questionnaire (CHQ)	QOL in children was similar to the normal range for healthy children	QOL is preserved in children with hepatitis B	7
Chen et al (Taiwan)	2017	Quantitative study	Cross-sectional	135 with hepatitis B or C	Hepatitis B or C: 49% men, mean age 56 years	QOL using chronic liver disease questionnaire (CLDQ) and short-form 36 (SF-36)	Mean CLDQ was 5.70, physical component summary score 45.94 and mental component summary score 49.37	QOL was poor, especially in those who needed education on the disease	6
Huang et al (China)	2018	Quantitative study	Cross-sectional	997 with hepatitis C	Hepatitis C: 55% men, median age 46 years	QOL using European Quality of Life scale (EQ-5D)	Mean EQ-5D score was 0.78 and EQ-VAS score was 77.2	Hepatitis C has an adverse effect on patients´ QOL. Depression,	8

								disease severity and symptom severity was associated with low QOL	
Horvath et al (Hungary)	2018	Quantitative study	Cross-sectional	60 with hepatitis C GP from population norms from questionnaire	Hepatitis C: 65% woman, mean age 53 years GP not described	QOL using short-from 36 (SF-36)	Patients had lower mean score in all domains compared to GP	QOL was significantly poorer than that of the GP. A negative correlation was found between the progression of cirrhosis and the physical domain of QOL	7
Jang et al (Korea)	2018	Quantitative study	Cross-sectional	235 with hepatitis C GP from population norms from questionnaire	Hepatitis C: 55% woman, mean age 61 years, 23% with cirrhosis GP not described	QOL using European Quality of Life scale (EQ-5D) and short-form 36 (SF-36)	Mean EQ-5D score was 0.85 and EQ-VAS score was 69.9. Physical component summary score 48.8 and mental component	Patients has lower QOL than GP. QOL was affected by the severity of disease, income, comorbidities, and medical response	7

							summary score 46.2		
Jang et al (Korea)	2019	Quantitative study	Cross-sectional	168 with hepatitis B	Hepatitis B: 68% men, mean age 49 years	QOL using Hepatitis B Quality of Life (HBQOL)	Mean overall score 64.41	QOL is impaired in patients with hepatitis B	8
Cossais et al (France)	2019	Quantitative study	Cross-sectional	505 with hepatitis C	Hepatitis C: 52% men, 72% over 50 years	QOL using European Quality of Life scale (EQ-5D)	Mean EQ-5D score was 0.80 and mean VAS score 68.8	Disease progression and comorbidities was associated with worse QOL	7
Daida et al (USA)	2020	Quantitative study	Survey from the Chronic Hepatitis Cohort	969 with hepatitis B or C	Hepatitis B or C: 51% men, mean age 49 years, 2% with cirrhosis	QOL using Short-Form 8 (SF-8)	Mental component score 51.3 Physical component score 51.0	Stressful life events and unemployment were related to poor QOL	6
Gupta et al (India)	2020	Quantitative study	Cross-sectional	150 with hepatitis B	Hepatitis B: 85% men, mean age 39 years, 8 % with cirrhosis	QOL using World Health Organization Quality of Life-Bref (WHO QoL-Bref) and CLDQ	General 4.8, physical health 19.6, psychological health 15.8, social 7.8, environmental 22.8 Weighted CLDQ score 23.7	QOL is impaired in patients with hepatitis	6
Zhang et al (China)	2020	Quantitative study	Cross-sectional	314 with hepatitis C	Hepatitis C: 69% woman,	QOL using European Quality	Compared with healthy controls,	Low QOL was associated with	7

				67 healthy controls	mean age 55 years. Healthy controls: 64% woman, mean age 55 years	of Life scale (EQ-5D) and chronic liver disease questionnaire (CLDQ)	patients with hepatitis C has greater negative impact on QOL.	female gender, comorbidity, drinking, and long disease duration	
Wallace et al (Australia)	2011	Qualitative study	Grounded theory	20 with hepatitis B 40 healthcare professionals (not included in the review)	Hepatitis B: 65% men, age between 19 and 60 years	Semistructured interview	Giving the diagnosis was shocking. Limited information and understanding created considerable confusion and fear for the future. Fear of rejection or stigma was identified as important. Try to manage the disease due to limited knowledge, beliefs and concerns	Patients need appropriate education and information, particularly at the point of diagnosis	7

Hill et al (England)	2014	Qualitative study	Phenomenological	23 with hepatitis C	Hepatitis C: 48% men, mean age 48 years	Unstructured interview	The diagnosis triggered a life transition with disappointment and lack of support and anxiety for progression. Stigma from other due to lack of knowledge. Try to positively handle the disease and treatment yet still living in uncertainty	Working within a chronic care framework of ongoing advice and support can improve experiences for those living with hepatitis	8
HassanpourDehkordi et al (Iran)	2016	Qualitative study	Constant-comparative thematic analysis	18 with hepatitis B or C	Hepatitis B or C: 83% men, mean age 37 years	Semistructured interview	Stigma was the main theme with sub-themes including fear to lose of family and social support, fear to present in public and fear of transmission	Stigma present a major challenge for patients with hepatitis	8

Wallace et al (China)	2017	Qualitative study	Thematic analysis	41 with hepatitis B	Hepatitis B: 71% men, 78% under 35 years	Semistructured interview	Hepatitis B is understood as an intergenerational family condition, that there are a substantial economic and physiological impact , disclosure was found to influence employment choices, and social marginalization was major issue	A national response to the infection needs to address the social as much as the clinical management and public health or preventive aspect	8
Adjei et al (Ghana)	2017	Qualitative study	Content analysis	14 with hepatitis B	Hepatitis B: 79% woman, age between 26-45 years	Semistructured interview	Patients were unclear about the impact of their infection. They experienced psychological and social problems especially when they were initially informed	The outcome suggests a public health intervention that can critically respond appropriately to this burden currently confronting	7

							about their disease. Sadness, fear, shock, shame and disbelief were reported. Coping strategies include religiosity, denial and lifestyle modification	patients with hepatitis	
Ezbarami et al (Iran)	2017	Qualitative study	Thematic analysis	27 with hepatitis B	Hepatitis B: 56% men, mean age 36	Semistructured interview	Data revealed that patients experiences insufficient self-care, misperceptions, stigmatization, psychological consequences, failure, spiritual struggle and post-traumatic growth	Capturing the experiences of hepatitis B patients can enhance the QOL of patients. Emotional support, unlabeled care and necessary information to patients can help patients adjust to hepatitis B	10
Valizadeh et al (Iran)	2019	Qualitative study	Content analysis	18 with hepatitis B	Hepatitis B: 56% men, mean age 35 years	Unstructured interview	Patients attempting an active expansion of interactions.	Consideration of education and consultation program regarding	8

							Patients adopted a number of strategies and approaches such as denial of the disease, protecting oneself or others to cope with the disease. Also religion	the different aspects of the disease is urgent	
Non-alcoholic fatty liver disease (NAFLD)									
Dan et al (USA)	2007	Quantitative study (questionnaire)	Cross-sectional	106 with NAFLD	NAFLD: 30% men mean age 46 years, 14% with cirrhosis. GP: 50% men, mean age 48 years.	QOL using chronic liver disease questionnaire (CLDQ)	The mean overall score was 5.4, abdominal symptoms 5.7, fatigue 4.8, systemic symptoms 5.5, activity 5.3, emotional functioning 5.4, worry 5.7	QOL is severely impaired in patients with NAFLD	8
David et al (USA)	2009	Quantitative study (questionnaire)	Cross-sectional from database with patient-	713 with NAFLD	NAFLD: 38% men, mean age	QOL using short-form 36 (SF-36)	Patients with NAFLD had worse physical and social	Patients with NAFLD have impaired QOL,	8

			reported outcomes	GP from population norms from questionnaire	48 years, 9 cirrhosis. GP not described.		functioning, role limitation due to physical and emotional health problems, general health perceptions, bodily pain and vitality than GP	especially those with severe NAFLD and increased fibrosis. Impairment is most evident in physical health while mental health is affected to a lesser degree	
Kistler et al (USA)	2010	Quantitative study (questionnaire)	Cross-sectional from database with patient-reported outcomes	239 children with NAFLD 5480 healthy children from reference sample	NAFLD: 72% boys, mean age 13 years Healthy children not described.	QOL using pediatric quality of life inventory (PedsQL)	Mean overall QOL score 72.7 in NAFLD children and 83.8 in healthy children	Children with NAFLD have a decrement in QOL. Symptoms were a major determinant of this impairment.	7
Kerkar et al (USA)	2013	Quantitative study (questionnaire)	Cross-sectional	48 children with NAFLD 40 obese control children	NAFLD: Mean age 12 years. Control group: Mean age 12 years	QOL using pediatric quality of life inventory (PedsQL)	Mean overall QOL score 74.58 in NAFLD children and 78.96 in obese control children	QOL in children with NAFLD was not significantly different from obese children without NAFLD	6
Golabi et al (USA)	2016	Quantitative study (questionnaire)	Cross-sectional National Health Survey	3333 with NAFLD 5982 GP	NAFLD: 58% men, mean age 51 years.	QOL using national health and nutrition	Patients with NAFLD rated there health worse than	NAFLD causes impairment of QOL	8

					GP: 50% men, mean age 47 years.	examination survey health-related quality of life -4 questionnaire (NHANES HRQOL-4)	GP, and patients with NAFLD had more day with poor physical and/or mental health than GP		
Tapper et al (USA)	2016	Quantitative study (questionnaire)	Prospective study	90 with NAFLD	NAFLD: 60% men, mean age 52 years	QOL using chronic liver disease questionnaire (CLDQ)	The mean overall score was 5.6, abdominal symptoms 6.0, fatigue 5.2, systemic symptoms 5.6, activity 6.0, emotional functioning 5.5, worry 5.5	QOL was the same as described in earlier studies. A 5% weight loss resulted in increased QOL	8
Sayiner et al (USA)	2016	Quantitative study (questionnaire)	Prospective study	89 with NAFLD GP from population norms from questionnaire	NAFLD: 44% men, mean age 51 years, 34% with cirrhosis. GP not described.	QOL using short-form 36 (SF-36)	Patients with NAFLD had worse physical and social functioning, role limitation due to physical and emotional health	QOL is impaired in patients with NAFLD	7

							problems, general health perceptions, bodily pain and vitality than GP		
Younossi et al (USA)	2017	Quantitative study (questionnaire)	Cross-sectional	25 with NAFLD	Not described	QOL using chronic liver disease questionnaire-non-alcoholic fatty liver disease (CLDQ-NAFLD)	The mean overall score was 5.6, abdominal symptoms 6.3, fatigue 4.8, systemic symptoms 5.5, activity 5.8, emotional functioning 5.6, worry 6.1		6
Huber et al (England, Germany, Spain)	2019	Quantitative study (questionnaire)	Prospective study	304 with NAFLD	NAFLD: 53% men, mean age 52 years.	QOL using chronic liver disease questionnaire (CLDQ)	The mean overall score was 4.99, abdominal symptoms 5.33, fatigue 4.31, systemic symptoms 5.09, activity 5.43, emotional	Patients had low QOL, especially those with comorbidities, who are older, woman and have more advanced disease	8

							functioning 4.93, worry 5.18		
Samala et al (USA)	2020	Quantitative study (questionnaire)	Prospective study	341 with NAFLD GP from population norms from questionnaire	NAFLD: 40% men, mean age 53 years. GP not described.	QOL using short-form 36 (SF-36)	Patients with NAFLD had worse physical and social functioning, role limitation due to physical and emotional health problems, general health perceptions, bodily pain and vitality than GP	Patients with NAFLD have lower QOL than GP. Body composition such as high body mass index and body fat was associated with worse QOL	8
Avery et al (England)	2017	Qualitative study	Content analysis	12 with NAFLD 21 healthcare professionals (not included in the systematic review)	Not described	Semistructured interview	Patients reported lack of information and knowledge on NAFLD and that it could be improved by lifestyle modification. Were advised to lose weight, but did not receive any support. Were advised that	Information and support at the time of diagnosis is severely lacking	7

							NAFLD was nothing to be concerned about		
Nonalcoholic steatohepatitis (NASH)									
Chawla et al (USA)	2016	Quantitative study (questionnaire)	Cross-sectional	79 with NASH GP from population norms from questionnaire	NASH: 35% men, mean age 46 years. GP not described.	QOL using chronic liver disease questionnaire (CLDQ) and short-form 36 (SF-36)	Patients with NAFLD had worse physical and social functioning, role limitation due to physical and emotional health problems, general health perceptions, bodily pain and vitality than GP. The mean overall score was 5.4, abdominal symptoms 5.5, fatigue 4.6, systemic symptoms 5.7, activity 5.8, emotional	Both QOL questionnaires revealed reductions in QOL	8

							functioning 5.7, worry 5.3		
Balp et al (Germany, France, Italy, Spain, England)	2019	Quantitative study (questionnaire)	Cross-sectional survey	184 with NASH 79267 GP cohort	NASH: 57% men, mean age 55 years. GP: 45% men, mean age 48 years	QOL using short-form 36 (SF-36), European Quality of Life scale (EQ-5D)	Mental component summary 39.22 Physical component summary 42.84 EQ-5D 0.67 in NASH. Mental component summary 45.16 Physical component summary 47.76 EQ-5D 0.78 in GP.	QOL is lower in patients compared to GP cohort. The burden of NASH may be underestimated.	8
Younossi et al (USA)	2019	Quantitative study (questionnaire)	Cross-sectional from database with patient-reported outcomes	1667 with NASH GP from population norms from questionnaire	NASH: 40% men, mean age 58 years, 52% with cirrhosis GP not described	QOL using chronic liver disease questionnaire (CLDQ) and short-form 36 (SF-36)	Patients with NAFLD had worse physical and social functioning, role limitation due to physical health problems, and bodily pain than GP.	QOL was lower in patients with NASH, especially those with advanced fibrosis and cirrhosis	8

							The mean overall score was 5.16, abdominal symptoms 5.38, fatigue 4.66, systemic symptoms 5.03, activity 5.43, emotional functioning 5.32, worry 5.12		
Cook et al (England, USA)	2019	Qualitative study	A combination of content analysis, grounded analysis, and discourse analysis	16 with NASH	NASH: 38% men.	Online qualitative forum for interactive discussion using different questions on patients perception of NASH	Patient with NASH have poor understanding of their disease, diagnosis of NASH is incidental, comorbidities are more concerning to patients than NASH, patients perceive that NASH impacts their social life and work performance in	Patients with NASH lack knowledge and have poor understanding of the disease and the symptoms commonly associated with the disease.	8

							more advanced stages		
Cirrhosis									
Marchesini et al (Italy)	2001	Quantitative study	Case-control	544 with cirrhosis 2031 controls	Cirrhosis: 64% men, mean age 60 years. Cause viral hepatitis 64%, alcohol 29%, 77% Child-Pugh A or B. Controls: 49% men, age > 45 years	QOL using short-form 36 (SF-36)	Patients with cirrhosis has poor QOL compared to controls	QOL is impaired in patients with cirrhosis. QOL was associated with distress, muscle cramps, and disease severity	7
Sumskiene et al (Lithuania)	2006	Quantitative study	Case-control	131 with cirrhosis 262 controls	Cirrhosis: 52% men, > 40 years 18%, 40-50 years 26%, 51-60 years 25%, < 60 years 31%. 79% Child-Pugh A or B. Cause 41% viral hepatitis, 38% alcohol.	QOL using chronic liver disease questionnaire (CLDQ)	The overall CLDQ score for patients with cirrhosis was lower than in persons with no cirrhosis (59,5 versus 85.3)	General QOL and QOL in all health domains were lower in patients with cirrhosis than in controls selected from the normal population. Increased in severity of disease	6

					Controls not described			was associated with poorer QOL.	
Kanwal et al (USA)	2009	Quantitative study	Cross-sectional	156 with cirrhosis	Cirrhosis: 55% men, mean age 54 years. Mean MELD 18.5.	QOL using the short form liver disease quality of life (SF-LDQOL)	The mean LDQOL score was 55.3. Physical component summary 30.8 Mental component summary 41.0	QOL is poor in patients with cirrhosis. Low QOL predicts higher mortality.	7
Les et al (Spain)	2010	Quantitative study	Cross-sectional	212 with cirrhosis	Cirrhosis: 74% men, mean age 62 years	QOL using chronic liver disease questionnaire (CLDQ) and short-form 36 (SF-36)	The mean CLDQ score was 4.8. Physical component summary 38.6 Mental component summary 45.3	QOL is impaired in patients with cirrhosis, especially those with disease severity, ascites, hepatic encephalopathy, decreased albumin and hemoglobin	7
Bajaj et al (USA)	2011	Quantitative study	Cross-sectional	100 with cirrhosis GP from population norms from questionnaire 100 relatives	Cirrhosis: 71% men, mean age 58 years GP not described	QOL using patients reported outcomes measurement information system (PROMIS) and sickness	Total SIP score 19.9	All patients scores were significantly impaired compared to GP	7

				(not included in systematic review)		impact profile (SIP)			
Parkash et al (Pakistan)	2012	Quantitative study	Cross-sectional	273 with cirrhosis	Cirrhosis: 57% men, mean age 49 years. 77% Child-Pugh A or B, mean MELD 12.6	QOL using chronic liver disease questionnaire (CLDQ)	The mean score was 4.36	Frequency of poor QOL is high in patients with cirrhosis.	7
Thiele et al (Denmark)	2013	Quantitative study	Prospective study	92 with cirrhosis	Cirrhosis: 54% men, mean age 61 years	QOL using chronic liver disease questionnaire (CLDQ)	Mean score was 4.4	Cirrhosis affect QOL. Low QOL was associated with body-mass index, etiology and severity of cirrhosis	8
Shiraki et al (Japan)	2013	Quantitative study	Cross-sectional	114 with cirrhosis GP from population	Cirrhosis: 56% men. 84% Child-Pugh A or B, 81% viral hepatitis. GP not described	QOL using short-form 36 (SF-36)	QOL in patients with cirrhosis was lower in al subscales than GP	QOL in patients with cirrhosis is low and correlated with disease severity	5
Zhang et al (China)	2016	Quantitative study	Cross-sectional	464 with cirrhosis	Cirrhosis: 61% men, mean age 56 years	QOL using liver cirrhosis patient-reported outcome	Total LC-PROM score 189 in patients with	Cirrhosis affect patients QOL	7

				112 healthy controls	Controls: 57% men, mean age 54 years	measure (LC-PROM)	cirrhosis and 241 in healthy controls		
Obradovic et al (Serbia)	2017	Quantitative study	Cross-sectional	15 with cirrhosis 15 healthy controls	Cirrhosis: 87% men, mean age 65 years. Mean Child-Pugh 6. Controls: 33% men, mean age 60 years	QOL using chronic liver disease questionnaire (CLDQ) and general health questionnaire (GHQ-12)	The mean overall score was 5.44 in patients with cirrhosis and 5.20 in healthy controls. Median GHQ-12 score 11 in patients with cirrhosis and 9 in healthy controls	Patients with cirrhosis has poorer QOL than healthy controls.	5
Janani et al (India)	2018	Quantitative study	Case-control	149 with cirrhosis 149 controls	Cirrhosis: 85% men, mean age 52 years. 62% Child-Pugh A or B, median MELD 24. Controls: 85% men, mean age 51 years	QOL using chronic liver disease questionnaire (CLDQ) and short-form 36 (SF-36)	Scores were significantly lower in patients with cirrhosis compared to controls	Patients with cirrhosis has poor QOL. This is affected by disease severity	6
Tapper et al (USA)	2019	Quantitative study	Cross-sectional	300 with cirrhosis	Cirrhosis: 56% men, mean age 60 years.	QOL using short-form 8 (SF-8)	Overall median SF-8 was 75	Disability, chair-stand performance,	7

					70% Child-Pugh A, mean MELD 9.			cognitive dysfunction, and psychoactive medication was associated with poor QOL	
Nagel et al (Germany)	2020	Quantitative study	Prospective study	42 with cirrhosis 21 with acute on chronic liver failure (not included in systematic review)	Cirrhosis: 40% men, mean age 58 years. 41% alcoholic, 16% NAFLD, mean MELD 15.	QOL using chronic liver disease questionnaire (CLDQ)	Overall mean QOL score was 4.33	Patients with decompensated cirrhosis suffer from impaired QOL	7
Hansen et al (USA)	2020	Quantitative study	Cross-sectional	66 with cirrhosis 66 relatives (not included in systematic review)	Cirrhosis: 63% men, mean age 56 years. 23% viral hepatitis, 32% alcohol, 29% NAFLD. Mean MELD 17.	QOL using short-form 36 (SF-36)	Physical summary score was 33.72 and mental health summary score was 42.74	Patients with cirrhosis has impaired QOL, especially those that are younger, has ascites or a history with hepatic encephalopathy	7
Abdi et al (Iran)	2015	Qualitative study	Phenomenological	10 with cirrhosis	Cirrhosis: 70% men, age	Semistructured interview	Patients felt stress and tension due to	Understanding the patients´	8

					between 39-54 years		physical condition, fear of disease outcome, and treatment cost. They express need for awareness, education and support to cope with the disease and reduce anxiety Patients experiences spirituality, which gave them faith and hope. They expressed that encouraged and effective communication result higher coping	experiences is a good method for demining appropriate treatment	
Shabanloei et al (Iran)	2017	Qualitative study	Content analysis	15 with cirrhosis	Cirrhosis: 47% men, mean age 46 years.	Semistructured interview	Patients felt disappointed because the disease was	The findings emphasize the necessity of psychological	

							discovered to late. Patients felt loss of self and difficulty with treatment compliance possibly due to lacking knowledge. A wish to die when facing complications and limited treatment options. Spiritual and religious issues affected patients	support for patients to cope with disease and treatment	
Fagerström et al (Sweden)	2017	Qualitative study	Inductive content analysis	13 with cirrhosis	Cirrhosis: 46% men, age between 46-70 years	Semistructured interview	Patients felt vulnerability when experiencing symptoms, loneliness, limits in daily living, and negative preconceptions. They accepted the disease and the effect on their	It is important to identify patients need for understanding and more knowledge and information to support and empower the patients	10

							situation. Patients felt responsible for observation of signs and symptoms, adherence, and asking for understanding and information		
Hepatocellular carcinoma (HCC)									
Steel et al (USA)	2007	Quantitative study (questionnaire)	Cross-sectional	83 with HCC 138 from GP 51 with CLD (not included in systematic review)	HCC: 77% men, mean age 58 years, 76% with cirrhosis. 93% Child-Pugh A or B. GP: 30% men, mean age 40 years.	QOL using Functional Assessment of Cancer Therapy-Hepatobiliary Carcinoma (FACT-Hep)	Patients with HCC had more additional concerns, poorer over-all QOL and emotional functional, and physical wellbeing than GP	Patients with HCC reports poorer QOL compared to GP	9
Sun et al (USA)	2008	Quantitative study (questionnaire)	Cross-sectional	22 with HCC 23 with pancreas cancer (not included in	HCC: 72% men, mean age 59 years.	QOL using Functional Assessment of Cancer Therapy-Hepatobiliary	The mean score for overall QOL was 110.6, the mean score physical well-being 14.9, social	Overall QOL is poor in patients with HCC. Specific QOL domains, such as physical and	8

				systematic review)		Carcinoma (FACT-Hep)	well-being 22.1, emotional well-being 14.8, functional well-being 13.5, and disease-specific symptoms 45.3	functional well-being, are of particular concern for patients.	
Ryu et al (Korea)	2010	Quantitative study (questionnaire)	Cross-sectional	180 with HCC	HCC: 89% men, mean age 55 years. 99% Child-Pugh A or B.	QOL using Functional Assessment of Cancer Therapy-Hepatobiliary Carcinoma (FACT-Hep)	The mean score for overall QOL was 126.48, with the mean score physical well-being 21.52, family well-being 16.85, emotional well-being 16.53, functional well-being 17.52, and functional assessment of cancer therapy 72.43	Patients with HCC has poor functional status and impaired QOL especially patients with more severe symptom experience	8
Qiao et al (China)	2012	Quantitative study (questionnaire)	Cross-sectional	140 with HCC	HCC: 95% men, mean age 52 years.	QOL using Functional Assessment of	The mean score for physical well-being 16.47, family well-	QOL of patients with HCC worsens gradually with	8

					82% Child-Pugh A or B.	Cancer Therapy-Hepatobiliary Carcinoma (FACT-Hep)	being 14.52, emotional well-being 12.74, functional well-being 17.57, and functional assessment of cancer therapy 41.46	progression of disease	
Fan et al (Taiwan)	2013	Quantitative study (questionnaire)	Cross-sectional	286 with HCC GP from population norms from questionnaire	HCC: 76% men, mean age 60 years. 93% Child-Pugh A or B. GP not described.	QOL using European Organization for Research and Treatment of Cancer quality of life questionnaire (EORTC QLQ-C30) and EORTC Quality of Life Questionnaire – Hepatocellular Carcinoma / Primary Liver Cancer Module	Patients with HCC had worse global QOL, physical, role limitation, cognitive, and social functioning, but better emotional functioning than GP	Patients with HCC had worse QOL compared to GP	9

						(EORTC QLQ-HCC18)			
Palmieri et al (Italy)	2015	Quantitative study (questionnaire)	Cross-sectional	24 with HCC 20 from GP 22 with cirrhosis (not included in systematic review)	HCC: 75% men, mean age 71 years. 100% Child-Pugh A or B. GP: 25% men, mean age 36 years.	QOL using short-form 36 (SF-36)	Patients with HCC had worse general mental health, psychological distress and well-being, and worse physical and social functioning, role limitation due to physical and emotional health problems, general health perceptions, and vitality than GP.	QOL is impaired in patients with HCC	8
Li et al (Hong Kong)	2019	Quantitative study (questionnaire)	Cross-sectional	472 with HCC	HCC: 89% men, mean age 60 years. 96% Child-Pugh A or B. MELD mean 9.02.	QOL using European Organization for Research and Treatment of Cancer quality of life questionnaire (EORTC QLQ-C30)	Global QOL 52.22, physical function 72.27, role function 74.61, emotional function 70.67, cognitive function 76.80,	In HCC patients, baseline QOL is low and correlated with liver function.	9

						and EORTC Quality of Life Questionnaire – Hepatocellular Carcinoma / Primary Liver Cancer Module (EORTC QLQ-HCC18)	social function 68.46, C30 index-score 30.69, HCC18 index-score 25.37		
Fan et al (Taiwan)	2012	Qualitative study	Phenomenological	33 with HCC	HCC: 67% men, mean age 54 years.	Semistructured interview	HCC was associated with physical symptoms and psychosocial stress, as well as positive changes as patients perceived HCC as a chronic disease that could be controlled and managed by focusing on symptoms emotional responses leading to a normal life.	Patients´ physical condition, their illness perceptions, and coping strategies contributed to their disease adjustment. Patients are keen for information about their disease.	10

Hansen et al (USA)	2015	Qualitative study	Conventional content analysis	14 with HCC	HCC: 79% men, mean age 61 years.	Semistructured interview	Patients wished to control how HCC was affecting there lives and quality of life by illness perception, decision to start treatment, and navigating treatment over time.	Healthcare professionals may improve patients' quality of life by eliciting patients´ perceptions of their illness and treatment decisions, symptoms experiences, and information needs as the disease progresses and providing symptom management and offering information tailored their needs.	9

AIH, autoimmune hepatitis; AILD, autoimmune liver disease; CLD, chronic liver disease; GP, general population; HCC, hepatocellular carcinoma; NAFLD, non-alcoholic fatty liver disease; NASH, non-alcoholic steatohepatitis; PBC, primary biliary cholangitis; QOL, quality of life